

AMBIDEXTROUS ONE-HAND TOUCH TYPING

Roadmap, training protocol, drills, benchmarks and reference material

Core target: QWERTY + one-hand Dvorak, independently with either hand

Project premise

Train one hand as a complete typing instrument, then make the skill ambidextrous. The goal is not merely speed: it is reliable touch typing, low visual dependence, controlled switching between motor maps, and a reproducible hobby experiment.

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Contents

1	Mission and scope
2	Mode taxonomy
3	What is actually standardized
4	Setup and ergonomics
5	Technique: purpose-built vs full-board
6	20-week core roadmap
7	Daily and weekly session design
8	Curriculum for the core layouts
9	Switching and interference training
10	Benchmarks and logging
11	Brain-health perspective
12	Tools and study material
13	Reference keymaps
14	Printable logs and checklists
15	Dual-machine split-task endgame

Rule of the project

Calendar weeks are suggestions. Progress gates matter more than dates. If accuracy is unstable, stay on the current mode; if a mode is already easy, advance early.

1. Mission and scope

The core end state is deliberately excessive but tightly scoped: you can sit at a normal QWERTY keyboard or select one-hand Dvorak, choose either hand, and type without looking down. QWERTY uses a full-board one-hand technique; Dvorak-L and Dvorak-R are purpose-built one-hand layouts. Colemak, Workman and other layouts are reserved for optional post-completion expansion.

Primary objectives

- Acquire four core one-hand modes: QL, QR, DL and DR.
- Maintain $\geq 98\%$ accuracy before chasing speed.
- Reduce visual keyboard checks to effectively zero.
- Build both dominant- and non-dominant-hand finger independence.
- Learn to switch layouts and hands without long reorientation periods.
- Keep the project measurable: WPM, accuracy, error profile, switch latency and fatigue.

Post-completion expansion modes

- Colemak Left / Right using the full-board one-hand technique.
- Workman Left / Right using the full-board one-hand technique.
- Half-QWERTY Left / Right (documented mirror-modifier technique).
- Further extras: Half-Colemak, Colemak-DH and other custom layouts after the core program is complete.

Success definition

A mode is considered "acquired" at 30 WPM, $\geq 98\%$ accuracy, no visual keyboard checking, and repeatable performance across at least three sessions. The long-term target can then move to 45-60+ WPM.

2. Mode taxonomy

Code	Layout	Hand	Type	Status
QL	QWERTY	Left	Full-board	Experimental technique
QR	QWERTY	Right	Full-board	Experimental technique
DL	Dvorak Left-Handed	Left	Purpose-built	Established OS layout
DR	Dvorak Right-Handed	Right	Purpose-built	Established OS layout

Why the distinction matters

The four core modes are not scientifically equivalent. DL/DR alter the logical layout specifically for one-hand use, while QL/QR keep ordinary QWERTY and ask one hand to cover the entire board. Treat speed comparisons as observations about complete systems. Colemak and Workman are intentionally excluded from the core comparison until the program is complete.

3. What is actually standardized

Dvorak Left-Handed / Right-Handed

These are real, purpose-built one-hand Dvorak variants. Apple documents both as selectable input sources, and the X Keyboard Extension (XKB) data used on Linux includes separate `dvorak-l` and `dvorak-r` definitions. [1][2]

Half-QWERTY

Half-QWERTY is a documented one-handed technique by Matias, MacKenzie and Buxton. The unused half is mirrored onto the active half while a modifier is held, preserving homologous finger movements from ordinary QWERTY. In experiments with existing touch typists, participants surpassed hunt-and-peck after roughly 3-4 hours and reached about half of their two-handed rate after about 8 hours. [3]

Colemak - optional post-completion

Colemak is outside the core program. After QL/QR/DL/DR are acquired, it can be added as an optional full-board one-hand challenge. Its official FAQ mentions an experimental Half-Colemak, but there is no generally recognized purpose-built Colemak-L/Colemak-R counterpart comparable to one-hand Dvorak. [4]

Workman - optional post-completion

Workman is outside the core program and is a potential expansion only after QWERTY and one-hand Dvorak are complete. It is a conventional alternative layout designed around physical finger travel and comfort; a standard one-hand left/right pair is not part of the main layout specification. [5]

Terminology in this guide

"Official" means a documented or widely distributed layout/technique. "Experimental" means this project imposes a consistent one-hand method on a normal two-hand layout.

4. Setup and ergonomics

Hardware

- Use a normal full-size, TKL or 75% keyboard first. Avoid changing physical keycaps; visual mismatch encourages true touch typing.
- A keyboard with clear F/J tactile bumps helps you relocate after whole-hand translations.
- Keep the keyboard low enough that the wrist can remain close to neutral. Move the forearm/hand as a unit for large lateral travel.
- For the one-hand full-board modes, temporarily moving the keyboard slightly toward the active hand is acceptable, but log it so comparisons stay fair.

Pain rule

This is skill practice, not endurance training. Stop or reduce load if you develop persistent pain, numbness, tingling or loss of coordination. Do not use fatigue as a progression metric. Frequent short sessions are preferable to marathon sessions.

Software

- Install/select the intended operating-system layout rather than mentally translating characters.
- Disable autocorrect and predictive text during benchmark sessions.
- Use the same text source, test duration and correction policy when comparing modes.
- Keep a visible mode label on screen (for example "DL") during early switching practice to reduce accidental cross-map learning.

5. Technique: purpose-built vs full-board

A. Dvorak-L / Dvorak-R

Follow the layout as a genuine one-hand system. OneHandTyper gives explicit home-position instruction and finger prompts for both variants. For example, its right-handed home position is A-E-H-T under index-middle-ring-pinky; the left-handed version uses the mirrored home-area logic. [6]

B. Full-board QWERTY

There is no single canonical fingering. Use a controlled translation protocol rather than stretching from one fixed home row. The central principle is: move the whole hand to a local four-key window, type with relaxed fingers, then relocate using tactile landmarks.

1. Choose a left zone, center zone and right zone on the physical keyboard. Practice moving between zones without looking.
2. Within each zone, keep finger assignments consistent. Do not let the same key randomly migrate between fingers from repetition to repetition.
3. Use the thumb primarily for Space. Use the nearest Shift when practical; if a shift combination causes twisting, use Sticky Keys or a remapped one-shot Shift during learning.
4. Return to a known tactile anchor (usually F or J) after a large jump until spatial memory becomes reliable.
5. Maintain a "fingering ledger" for awkward keys (especially T/Y/G/H/B/N, brackets, Backspace and punctuation). Once a mapping is comfortable and accurate, freeze it for at least two weeks before reconsidering.

Do not confuse "one hand" with "one finger"

The goal is still touch typing: all available fingers participate. Large movements should come from hand/forearm translation, not repeated extreme finger extension.

Experimental fingering ledger

Key / position	Left-hand finger	Right-hand finger	Notes
T / G / B			
Y / H / N			
Backspace			
Enter			
Left Shift			
Right Shift			
[] \			
;/ ?			

6. 20-week core roadmap

Recommended load: 5 practice days per week, 30-45 minutes per day. Each four-week block has one primary core mode, followed by a four-week integration block. Previously acquired modes receive maintenance so early skills do not decay. Optional layouts begin only after the core completion gate.

Weeks	Primary	Maintenance	Gate before moving on
1-4	QL	Two-hand QWERTY baseline	20 WPM / 97% / no looking

5-8	QR	QL 2x/week	20 WPM / 97% / no looking
9-12	DL	QL + QR	20 WPM / 97%
13-16	DR	DL + Q pair	20 WPM / 97%
17-20	Core integration	All four	30 WPM / 98% on each; then optional extras may begin

If you want a faster, more chaotic version

Train both hands of the same system in parallel (for example QL on Monday/Wednesday, QR on Tuesday/Thursday, mixed on Friday; later repeat with DL/DR). This may improve symmetry but increases interference. The sequential plan above is the safer default for building four distinct core maps.

Maintenance rule

Once a mode reaches 30 WPM / 98%, one 5-10 minute session per week is enough initially. If its rolling 3-session average drops more than 15%, temporarily increase maintenance frequency.

7. Daily and weekly session design

30-minute standard session

1. 3 min - relocation warm-up: find home/anchors without looking; slow rows and simple alternations.
2. 7 min - weak-key isolation: only the current problem keys or zones.
3. 10 min - word practice: accuracy-first, unlimited time.
4. 7 min - timed text: 30-60 second tests with normal English.
5. 3 min - log results and one sentence about the biggest failure mode.

45-minute deep session

- 5 min warm-up.
- 10 min key/zone drills.
- 10 min common words and bigrams.
- 10 min prose or code.
- 5 min switch drill.
- 5 min benchmark + notes.

Weekly cadence

Day	Focus
Mon	Primary mode - accuracy
Tue	Primary mode - weak keys
Wed	Primary mode + one maintenance mode
Thu	Primary mode - prose/code
Fri	Benchmark + switching
Sat	Optional fun session / race / unusual text
Sun	Off or 5-minute effortless maintenance

8. Curriculum for the core layouts

Use the same skill ladder for every mode. The content changes, but the acquisition order stays stable. This makes comparisons cleaner.

Level 0 - Spatial map	Name/find every letter without looking. No speed target.
Level 1 - Home & anchors	Home-row patterns, Space, Backspace, Enter, Shift. $\geq 95\%$ accuracy.
Level 2 - Rows	Add top row, then bottom row. Short artificial sequences are acceptable.
Level 3 - Words	High-frequency English words; 10-15 WPM at $\geq 96\%$.
Level 4 - Prose	Normal sentences and punctuation; 20 WPM at $\geq 97\%$.
Level 5 - Acquired	30 WPM at $\geq 98\%$, no visual checking.
Level 6 - Strong	45 WPM at $\geq 98.5\%$ across three sessions.
Level 7 - Showcase	60+ WPM at $\geq 99\%$ or a personally chosen advanced target.

QWERTY

Use it to learn the full-board movement technique before introducing extra logical-map complexity. Because you already know where QWERTY characters live, failures are mostly motor/geometry failures rather than memory failures.

Dvorak-L/R

Treat these as new layouts, not as remapped QWERTY. Start with the dedicated tutor and home positions. Expect the layout-memory burden to be larger but the physical one-hand design to be more coherent.

Colemak - optional post-completion

After core completion: Colemak can be added as a full-board one-hand expansion. Because it preserves many QWERTY positions while moving others, expect "almost-QWERTY" interference and reintroduce explicit mode cues.

Workman - optional post-completion

After core completion: Workman can be added as another full-board one-hand expansion. Treat it as its own map while preserving the same physical movement protocol used for QWERTY.

9. Switching and interference training

Do not start with random switching

Random switching is the endgame. Early practice should create strong context cues: layout name on screen, a fixed block of time, and a deliberate 5-10 second reset before the first word.

Three-stage switching ladder

1. Blocked: 10-20 minutes in one mode before changing.
2. Paired: alternate two established modes every 2-5 minutes.
3. Randomized: a random code (QL/QR/DL/DR) appears before each 30-60 second trial.

Measure switch latency

Start a timer when the new mode cue appears. Stop it after the first 10 correct characters without correction. Track median latency. A reasonable hobby target is <10 seconds between familiar modes, eventually <5 seconds for well-trained pairs.

Interference is data

If QWERTY intrudes into Dvorak (or vice versa), or one hand "copies" the other hand, record the specific keys. Those errors reveal which representations are competing. Do not merely count them as generic mistakes.

10. Benchmarks and logging

Core metrics

Metric	Why it matters
WPM	Throughput; use only with accuracy.
Accuracy	Primary quality gate.
Raw WPM	Useful for separating hesitation from correction cost.
Visual glances	Should trend toward zero.
Switch latency	Measures motor-map selection.
Key-level errors	Identifies interference and awkward geometry.
Fatigue 0-5	Prevents speed gains from hiding ergonomic problems.
Session RPE 0-5	Subjective effort; useful when comparing layouts.

Benchmark protocol

- Use 60-second English tests for the main benchmark.
- Run three tests; record the median WPM and median accuracy.
- Do not benchmark immediately after a long fatigue session.
- Once per month, repeat the same fixed quotation or passage in every acquired core mode.
- For fair comparison, keep keyboard, chair, text source, correction policy and hand placement constant.

11. Brain-health perspective

This project is cognitively rich motor learning, but it should not be marketed to yourself as a medical intervention. Motor-skill learning is associated with functional and structural neuroplastic changes during acquisition and consolidation. [7] The four-mode core project additionally taxes procedural memory, fine motor control, error correction, inhibition of competing maps and task switching.

What is reasonable to expect

- Better skill at the trained tasks.
- Improved non-dominant-hand control and finger independence.
- Practice selecting and suppressing competing motor programs.
- Sustained novelty if you keep adding difficult but achievable variations.

What is not established

- That learning many keyboard layouts meaningfully raises general intelligence.
- That this specific hobby prevents dementia or neurological disease.
- That more layouts automatically produce broader cognitive transfer.

Brain-health framing

Think of it as a demanding lifelong skill hobby that contributes cognitive and motor stimulation. For general brain health, it complements rather than replaces exercise, sleep, cardiovascular risk management, social engagement and other learning.

12. Tools and study material

Monkeytype

Best general lab for the core project. During the program, use QWERTY and Dvorak-L/R; Colemak, Workman, Colemak-DH and other layouts are optional post-completion extras. <https://monkeytype.com/settings>

OneHandTyper

Dedicated left- and right-handed Dvorak tutor with home-position and finger prompts. <https://onehandtyper.com/>

York University / MacKenzie archive

Half-QWERTY papers and historical one-hand typing research. <https://www.yorku.ca/mack/hci96.html>

Colemak FAQ

Official Colemak background; includes note about experimental Half-Colemak. <https://colemak.com/FAQ>

Workman project

Layout philosophy, diagrams and implementation links. <https://workmanlayout.org/>

Apple Dvorak guide

Confirms selectable Dvorak Left-Handed and Dvorak Right-Handed layouts on macOS. <https://support.apple.com/guide/mac-help/mh27976/mac>

Suggested Monkeytype configuration

- 30-60 second tests during acquisition; 60 seconds for benchmark.
- Accuracy first; use stop-on-error sparingly so it does not destroy rhythm.
- Use keymap display only during early learning, then remove it.
- Use layout filters / weakspot-style practice to isolate problem keys.
- Use Dvorak-L and Dvorak-R layout selections rather than mentally remapping standard Dvorak.

13. Reference keymaps

QWERTY (letters/punctuation)

```
Q W E R T   Y U I O P
A S D F G   H J K L ;
Z X C V B   N M , . /
```

Dvorak - Left-Handed (XKB mapping)

```
[ ] / P F M L J 4 3 2 1
; Q B Y U R S O . 6 5 =
- K C D T H E A Z 8 7
```

' X G V W N I , 0 9

Dvorak - Right-Handed (XKB mapping)

1 2 3 4 J L M F P / []
 5 6 Q . 0 R S U Y B ; =
 7 8 Z A E H T D C K -
 9 0 X , I N W V G '

Colemak - optional post-completion

Q W F P G J L U Y ;
 A R S T D H N E I O
 Z X C V B K M , . /

Workman - optional post-completion

Q D R W B J F U P ;
 A S H T G Y N E O I
 Z X M C V K L , . /

Note: shown as logical character rows on a US-style physical keyboard. Exact OS punctuation/dead-key behavior can vary with locale and international variants.

14. Printable logs and checklists

Session log

Date	Mode	Minutes	Median WPM	Accuracy	Glances	Fatigue 0-5	Main error

Mode acquisition checklist

- ☐ Can find every letter without looking.
- ☐ Can type Space, Backspace, Enter and Shift comfortably.
- ☐ 10 WPM at >=95% accuracy.
- ☐ 20 WPM at >=97% accuracy.

- [] 30 WPM at >=98% accuracy in three sessions.
- [] No visual keyboard checking in normal prose.
- [] Can resume after a 1-week break with <15% speed loss.
- [] Can switch into the mode from another established mode in <10 seconds.

Monthly four-mode core benchmark

Mode	WPM	Accuracy	Switch latency	Fatigue	Notes
QL					
QR					
DL					
DR					

15. Dual-machine split-task endgame

What it is

Advanced optional end state: type on two different machines at once, one hand per machine, potentially with different layouts and different texts. This extends one-hand typing into dual-stream bimanual work.

Practical setup

- If possible, use different keyboard feel or placement for the two sides: different machines, different angles, or clearly separated left/right screens.
- Different core layouts can help by creating stronger context cues (for example left = QWERTY, right = Dvorak-R).
- During early sessions, keep the tasks simple and strongly separated: fixed strings, copy text, then copy text plus composition.
- Use visible mode labels, and if helpful use different wallpapers, text colors or editor themes until the contexts stop colliding.

Difficulty ladder

1. The same simple string on both machines.
2. Two different predetermined strings (for example letters on one side, numbers or symbols on the other).
3. Two different repeated phrases or short paragraphs.
4. Two different copy texts, one per machine/screen.
5. Copying on one side while freely composing on the other.
6. Two simultaneous dictation streams or two independent composition streams. This is a showcase target, not a near-term expectation.

Measuring performance

- Record solo baselines first: each hand, each layout, one text stream.
- In dual sessions, log left WPM, right WPM, combined throughput, cross-stream substitutions, stalls and attention collapses.
- A simple comparison metric is dual efficiency: $(\text{dual left WPM} + \text{dual right WPM}) / (\text{solo left WPM} + \text{solo right WPM})$.
- A useful long-term milestone is when combined dual-stream throughput exceeds your normal two-hand typing speed, even though the outputs are separate documents.

Cautions and interpretation

This is an optional showcase layer after the four core modes are stable. The likely bottleneck is cognitive rather than purely motor: copying two visible texts is much more tractable than composing two unrelated texts at the same time. Treat it as a divided-attention and context-separation experiment, not as a required part of the base roadmap.

A. Sources and notes

[1] **Apple Support** - Use Dvorak keyboard layouts on Mac - includes Dvorak Left-Handed and Right-Handed.

<https://support.apple.com/guide/mac-help/mh27976/mac>

[2] **XKB / libxkbcommon** - US symbol definitions include dvorak-l and dvorak-r keymaps.

<https://github.com/xkbcommon/libxkbcommon/blob/master/test/data/symbols/us>

[3] **Matias, MacKenzie & Buxton (1996)** - One-Handed Touch-Typing on a QWERTY Keyboard; Half-QWERTY training results and method. <https://www.yorku.ca/mack/hci96.html>

[4] **Colemak FAQ** - Colemak design/learning notes and experimental Half-Colemak mention. <https://colemak.com/FAQ>

[5] **Workman** - Official Workman layout philosophy and reference. <https://workmanlayout.org/>

[6] **OneHandTyper** - Interactive one-handed Dvorak tutor and home-position instructions. <https://onehandtyper.com/>

[7] **Dayan & Cohen (2011)** - Neuroplasticity subserving motor skill learning - review of functional and structural plasticity.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC3217208/>

Evidence note

The training schedule, proficiency gates, switching drills and full-board one-hand protocol are practical recommendations for this hobby project, not validated clinical or ergonomic prescriptions. The cited sources support the established layouts/techniques and the general motor-learning discussion.

END STATE: either hand + any core layout + no looking + controlled switching